Supporting information

Table of contents

JSON building blocks 5
Conventions used in the document 5
The top-level section 5
Reference data 6
Key performance metrics 6
Certified performance 7
Device classification 8
Device classification – top level 8
Device area 8
Band gap 8
Device stack 9
Device stack – top level 9
Layers 9
Air gap 9
Anti reflection 10
Back contact 10
Buffer layer 10
Edge sealing 11
Electrolyte 11
Encapsulation 11
Electron transport layer 11
Front contact 12
Hole transport layer 12
Interface modifier 12
Mesoporous scaffold 13
Middle contact 13
Optical_spacer 13
Other 14
Recombination_layer 14
Refractive index matching 14
Substrate 15
TCO 15
Layers photoabsorbers 16
CIGS 16
CIS 16

CZTS 16
OPV blend 16
Organic dye 17
Other 17
Quantum dots 18
Perovskite 18
silicon 19
Materials 20
Chemical 20
Chemical composition 20
Ink 20
Ligands 20
Materials in layer 20
Nanstructuration 21
Photoabsorber compositions 21
Composition CIGS 21
Composition CIS 21
Composition CZTS 21
Composition Perovskite 21
Perovskite ions 22
Secondary Phases Impurities and Dopands 22
Solution 22
Supplier 22
Target material 23
Deposition procedure 24
Cleaning procedure 24
Chemical bath deposition 24
Dip coating 24
Doctorblading 24
Etching procedure 25
Evaporation multi source 25
Evaporation single source 25
Heating 25
Inkjet printing 25
Ion exchange by dipping 26
Ion exchange by gas diffusion 26
Post deposition procedure 26
Slot dye coating 26

Spin-coating 26
Spray-coating 27
Sputtering 27
Storage 27
Storage until use 27
Deposition procedure – sub sections 29
Antisolvent details 29
Environmental conditions 29
Evaporation source 29
Evaporation segment 29
Gas quenching details 29
Spin-coating segments 29
Sputtering segments 30
Temperature segments 30
Measurements 31
Measurements - Top level 31
EQE 31
Flexibility 31
JV measurement 31
Outdoor performance 32
Stabilised performance 33
Stability 33
Transmission 34
Measurements – Sub sections 35
Certified performance 35
Certification details 35
EQE metrics 35
JV metrics 35
Light conditions 35
Load cycling 36
Outdoor metrics 36
Preconditioning 37
Stabilised performance metrics 37
Stability metrics 37
Measurements - Raw data 38
Raw data eqe 38
Raw data jv 38
Raw data outdoor performance 38

Raw data stabilized performance 38
Raw data stability 38
Raw data transmission 38
Module Data 39
Module data – top level 39
Module description 39

JSON building blocks

This document contains a description of the JSON schemas for all modules and submodules that are implemented in the project at the time of publication. This provides an overview of the format, key-value pairs, and preferred units when applicable. Most entries should be self-explanatory, but additional comments are given when warranted.

The data in this file should not be seen as the official version of the data schemas. The official versions of the data schemas, which may be updated and extended as long as the project is alive are available at **XXX.** Those are the versions being implemented in the Nomad system. The files are there stored as .yaml files with additional functionality for interfacing with the graphical user interface in the Nomad system.

Conventions used in the document

To simplify the reading of the data schemas, they have been color coded.

- Sections and subsections are given in purple.
- Keys are given in dark blue.
- Values are given in red.
- In the examples values are replaced with Strings indicating the datatype and the unit of the value as: as "Key": "Value [unit]"
- Values that have the form of lists with several values are indicated as "Key": "['*'] [unit]"
- Subsections are given by the name followed by empty angular brackets as: "subsection": "{}"
- Subsections that contain lists of entries are given as the name followed by a bracket with angular brackets as: "subsection": [{}, {}, …]
- Some fields are free text fields, but where there is a limited, but unspecified, number of valid alternatives. In those cases, we have listed common alternatives in brackets {}. Those alternatives are provided in dropdown menus in the graphical user interface

The top-level section

The top level is named data. It contains one key-value pair, and seven subsections.
The single key-value pair is for the database id the file gets when uploaded to an online repository.

```
{
    "data":{
        "database_id": "Integer"
        "reference_data": "{}",
        "key_performance_metrics": "{}",
        "certified_performance": "{}",
        "device_classification": "{}",
        "device_stack": "{}",
        "measurements": "[{}, {}, ...]",
        "module_data": "{}"
    }
}
```

Reference data

The reference data module has in its base occurrences eight data fields.

```
"reference_data": {
    "data_entered_by_author": "Boolean",
    "doi": "String",
    "free_text_comment": "String",
    "journal": "String",
    "lead_author": "String",
    "name_of_person_entering_the_data": "String",
    "publication_date": "Datetime",
    "sample_id": "String",
}
```

doi: is the DOI number of the publication where the original data is published. If a DOI number is provided, the Nomad back end will from Crossref programmatically access the publication_date, the journal, and the lead_autor.

sample_id: is the sample id the device have in the original publication.

Key performance metrics

This section contains the standard key metrics for the device. The assumption here is that measurements have been done under standard test conditions under AM 1.5 illumination. This should be the metrics for the complete device. In case of a four terminal device where there may be performance data foreach sub cell, data for those would be placed in the "measurement" module.

All fields will not be relevant for all devices. A 4-terminal device does for example have pce, but not a fill factor for the complete device.

There are no stability values given here, as there is not one single standard for measuring stability. Stability data is instead given under "measurements"

```
"key_performance_metrics": {
    "ff": "Float [j_mp*v_mp/(j_sc*voc)]",
    "j_sc": "Float [mA/cm²]",
    "j_mp": "Float [mA/cm²]"
    "pce_stabilized": "Float",
    "pce_from_iv_data": "Float",
    "pce_1000h_ISOS_L_1": "Float",
    "pce_1000h_ISOS_L_3": "Float",
    "T80_ISOS_L_1": "Float [hr]",
    "T80_ISOS_L_3": "Float [hr]",
    "v_mp": "Float [V]",
    "voc": "Float [V]",
}
```

Certified performance

The certified performance module contains a Boolean field indicating is the cell has been certified at an external certification institute. The performance metrics are the same as for the “key performance metrics”

```
"certified_performance": {
    "certification_date": "Datetime",
    "certification_institute": "String",
    "certification_protocol": "String",
    "ff": "Float [j_mp*v_mp/(j_sc*voc)]",
    "is_certified": "Boolean",
    "j_sc": "Float [mA/cm²]",
    "j_mp": "Float [mA/cm²]"
    "pce_stabilized": "Float",
    "pce_from_iv_data": "Float",
    "pce_1000h_ISOS_L_1": "Float",
    "pce_1000h_ISOS_L_3": "Float",
    "T80_ISOS_L_1": "Float [hr]",
    "T80_ISOS_L_3": "Float [hr]",
    "v_mp": "Float [V]",
    "voc": "Float [V]",
}
```

Device classification

This section contains one field called “band_gaps”. This is supposed to be a list of the band gaps of the photoabsorbers, starting from the lowest value, i.e. the photoabsorber in the bottom cell.

Device classification – top level

```
"device_classification": {
    "band_gaps": "['*'] [eV]",
    "is_bifacial": "Boolean",
    "is_encapsulated": "Boolean",
    "is_flexible": "Boolean",
    "is_semitransparent": "Boolean",
    "number_of_junctions": "Integer",
    "number_of_terminals": "Integer",
    "tandem_architecture": "String {Monolithic, Stacked, Spectral_splitter,
                                    Wide_bandgap_cell_used_as_reflector, ...}",
    "tandem_technology": "String {CIGS-Perovskite, CIGS-Perovskite-Perovskite, CIS-
                                Perovskite, CZTS-Perovskite, GaAs-Perovskite, OPV-
                                OPV-Perovskite, OPV-Perovskite, PbS-QD-Perovskite,
                                Perovskite-DSSC, Perovskite-OPV, Perovskite-PbS-
                                QD, Perovskite-Perovskite, Perovskite-Perovskite-
                                Perovskite, Silicon-Perovskite,  Silicon-
                                Perovskite-Perovskite, ...}",
    "device_area": "{}"
}
```

Device area

```
"device_area": {
    "cell_area": "Float [cm²]",
    "illuminated_cell_area": "Float [cm²]",
    "number_of_cells_per_substrate": "Integer",
    "substrate_area": "Float [cm²]",
}
```

Band gap

```
"Band_gap":{
    "estimated_from": "String {Absorption_onset, Tauc_analysis,
                            Absorption_inflection_point, EQE_onset,
                            EQE_inflection_point, Literature_value, Other}",
    "is_graded": "Boolean",
    "value": "Float [eV]",
}
```

Device stack

The central part of the device stack module is a subsection that list all the layers in the device stack. The two key-values stack_sequence and number_of_layers could be extracted programmatically from the list of layers provided.

The first key-value for each layer is the functionality of the layer in the device stack. Depending on functionality, different parameters will be of importance. We have therefore defined subsections for different functionalities. In the graphical user interface, the user selects a functionality whereupon ley-values and subsection relevant for a layer of that functionality is loaded level of nesting.

Each layer in the stack should be given a "layer_index", counting from one and up, which represent the layers position relative to other layers in the stack. With this index, the full device stack can simply be derived from the list of layers.

Earth layer should also specify the index of the layer used as a substrate ("layer_used_as_substrate") during deposition. This is required to extract the full synthetic protocol of the cell, as it from the layer stack alone not always is possible to deduce which layer that was deposited on which layer.

This indexing requires an agreement to what should be considered up and down in the device. Our proposition is that the convention should be that the layers should be ordered from left to right, under the assumption that light enters the device from the right. This is an arbitrary choice, but to obtain a coherent description a choice needs to be made. The general principle chosen can be summarised in the catchphrase "***Light from the right!***"

Device stack – top level

```
"device_stack":{
    "device_architecture": "String {Monolithic_device, 2_separable_subcells,
                                    3_separable_subcells, Other}",
    "stack_sequence": "String",
    "number_of_layers": "String",
    "layers": "[{}, {}, ...]"
}
```

Layers

```
"layers": [
    {
    "functionality": "String/Section {Air_gap, Anti_reflection, Back_contact,
                            Basic_layer, Buffer_layer, Edge_sealing, Electrolyte,
                            Encapsulation, ETL, Front_contact, HTL,
                            Interface_modifier, Middle_contact, Optical_spacer,
                            Other, Photoabsorber, Recombination_layer,
                            Refractive_index_matching, Substrate, TCO}"
    }
]
```

Air gap

```
{
    "functionality": "air_gap",
    "layer_index": "Integer",
    "layer_used_as_substrate": "Integer",
    "subcell_association": "String {Monolithic_device, Bottom_cell, Middle_cell,
                                    Top_cell, Other}",
    "thickness": "Float [mm]",
```

```
}
```

Anti reflection

```
{
    "functionality": "anti_reflection",
    "area": "Float [cm²]",
    "layer_index": "Integer",
    "layer_used_as_substrate": "Integer",
    "sheet_resistance": "Float [ohm]",
    "subcell_association": "String {Monolithic_device, Bottom_cell, Middle_cell,
                                    Top_cell, Other}",
    "surface_roughness": " Float [nm]",
    "thickness": "Float [nm]",
    "deposition_procedure": "[{}, {}, ...]",
    "materials_in_layer": "[{}, {}, ...]",
    "post_deposition_procedure": "[{}, {}, ...]",
    "storage_until_use": "{}"
}
```

Back contact

```
{
    "functionality": "back_contact",
    "area": "Float [cm²]",
    "deposition_location": "String {in_house, by_commercial_supplier, other}",
    "layer_index": "Integer",
    "layer_used_as_substrate": "Integer",
    "thickness": "Float [nm]",
    "sheet_resistance": "Float [ohm]",
    "subcell_association": "String {Monolithic_device, Bottom_cell, Middle_cell,
                                    Top_cell, Other}",
    "surface_roughness": " Float [nm]",
    "deposition_procedure": "[{}, {}, ...]",
    "materials_in_layer": "[{}, {}, ...]",
    "post_deposition_procedure": "[{}, {}, ...]",
    "storage_until_use": "{}"
}
```

Buffer layer

```
{
    "functionality": "buffer_layer",
    "area": "Float [cm²]",
    "deposition_location": "String {in_house, by_commercial_supplier, other}",
    "layer_index": "Integer",
    "layer_used_as_substrate": "Integer",
    "thickness": "Float [nm]",
    "sheet_resistance": "Float [ohm]",
    "subcell_association": "String {Monolithic_device, Bottom_cell, Middle_cell,
                                    Top_cell, Other}",
    "surface_roughness": " Float [nm]",
    "deposition_procedure": "[{}, {}, ...]",
```

```
    "materials_in_layer": "[{}, {}, ...]",
    "post_deposition_procedure": "[{}, {}, ...]",
    "storage_until_use": "{}"
}
```

Edge sealing

```
{
    "functionality": "edge_sealing",
    "subcell_association": "String {Monolithic_device, Bottom_cell, Middle_cell,
                                    Top_cell, Other}",
    "thickness": "Float [nm]",
    "water_vapour_transmission_rate": "Float [g/m**2/hr]",
    "oxygen_vapour_transmission_rate": "Float [cm**3/m**2/hr]",
    "deposition_procedure": "[{}, {}, ...]",
    "materials_in_layer": "[{}, {}, ...]",
}
```

Electrolyte

```
{
    "functionality": "electrolyte",
    "area": "Float [cm²]",
    "layer_index": "Integer",
    "layer_used_as_substrate": "Integer",

    "subcell_association": "String {Monolithic_device, Bottom_cell, Middle_cell,
                                    Top_cell, Other}",
    "thickness": "Float [nm]",
    "solution": "{}"
}
```

Encapsulation

```
{
    "functionality": "encapsulation",
    "area": "Float [cm²]",
    "layer_index": "Integer",
    "layer_used_as_substrate": "Integer",
    "oxygen_vapour_transmission_rate": "Float [cm**3/m**2/hr]",
    "subcell_association": "String {Monolithic_device, Bottom_cell, Middle_cell,
                                    Top_cell, Other}",
    "thickness": "Float [mm]",
    "water_vapour_transmission_rate": "Float [g/m**2/hr]",
    "deposition_procedure": "[{}, {}, ...]",
    "materials_in_layer": "[{}, {}, ...]",
}
```

Electron transport layer

```
{
    "functionality": "electron_transport_layer",
    "area": "Float [cm²]",
    "electron_mobility": "Float [cm²/(Vs)]",
    "layer_index": "Integer",
```

```
    "layer_used_as_substrate": "Integer",
    "sheet_resistance": "Float [ohm]",
    "surface_roughness": " Float [nm]",
    "subcell_association": "String {Monolithic_device, Bottom_cell, Middle_cell,
                                Top_cell, Other}",
    "thickness": "Float [nm]",
    "deposition_procedure": "[{}, {}, ...]",
    "materials_in_layer": "[{}, {}, ...]",
    "post_deposition_procedure": "[{}, {}, ...]",
    "storage_until_use": "{}"
}
```

Front contact

```
{
    "functionality": "front_contact",
    "area": "Float [cm²]",
    "layer_index": "Integer",
    "layer_used_as_substrate": "Integer",
    "sheet_resistance": "Float [ohm]",
    "surface_roughness": " Float [nm]",
    "subcell_association": "String {Monolithic_device, Bottom_cell, Middle_cell,
                                Top_cell, Other}",
    "thickness": "Float [nm]",
    "deposition_procedure": "[{}, {}, ...]",
    "materials_in_layer": "[{}, {}, ...]",
    "post_deposition_procedure": "[{}, {}, ...]",
    "storage_until_use": "{}"
}
```

Hole transport layer

```
{
    "functionality": "hole_transport_layer",
    "area": "Float [cm²]",
    "hole_mobility": "Float [cm²/(Vs)]",
    "layer_index": "Integer",
    "layer_used_as_substrate": "Integer",
    "sheet_resistance": "Float [ohm]",
    "surface_roughness": " Float [nm]",
    "subcell_association": "String {Monolithic_device, Bottom_cell, Middle_cell,
                                Top_cell, Other}",
    "thickness": "Float [nm]",
    "deposition_procedure": "[{}, {}, ...]",
    "materials_in_layer": "[{}, {}, ...]",
    "post_deposition_procedure": "[{}, {}, ...]",
    "storage_until_use": "{}"
}
```

Interface modifier

```
{
    "functionality": "interface_modifier",
    "area": "Float [cm²]",
```

```
    "layer_index": "Integer",
    "layer_used_as_substrate": "Integer",
    "sheet_resistance": "Float [ohm]",
    "surface_roughness": " Float [nm]",
    "subcell_association": "String {Monolithic_device, Bottom_cell, Middle_cell,
                                Top_cell, Other}",
    "thickness": "Float [nm]",
    "deposition_procedure": "[{}, {}, ...]",
    "materials_in_layer": "[{}, {}, ...]",
    "post_deposition_procedure": "[{}, {}, ...]",
    "storage_until_use": "{}"
}
```

Mesoporous scaffold

```
{
    "functionality": "mesoporous_scaffold",
    "area": "Float [cm²]",
    "layer_index": "Integer",
    "layer_used_as_substrate": "Integer",
    "subcell_association": "String {Monolithic_device, Bottom_cell, Middle_cell,
                                Top_cell, Other}",
    "thickness": "Float [nm]",
    "deposition_procedure": "[{}, {}, ...]",
    "materials_in_layer": "[{}, {}, ...]",
    "post_deposition_procedure": "[{}, {}, ...]",
    "storage_until_use": "{}"
}
```

Middle contact

```
{
    "functionality": "middle_contact",
    "area": "Float [cm²]",
    "deposition_location": "String {in_house, by_commercial_supplier, other}",
    "layer_index": "Integer",
    "layer_used_as_substrate": "Integer",
    "sheet_resistance": "Float [ohm]",
    "surface_roughness": " Float [nm]",
    "subcell_association": "String {Monolithic_device, Bottom_cell, Middle_cell,
                                Top_cell, Other}",
    "thickness": "Float [nm]",
    "deposition_procedure": "[{}, {}, ...]",
    "materials_in_layer": "[{}, {}, ...]",
    "post_deposition_procedure": "[{}, {}, ...]",
    "storage_until_use": "{}"
}
```

Optical_spacer

```
{
    "functionality": "optical_spacer",
    "area": "Float [cm²]",
    "layer_index": "Integer",
```

```
    "layer_used_as_substrate": "Integer",
    "surface_roughness": "Surface roughness [nm]",
    "subcell_association": "String {Monolithic_device, Bottom_cell, Middle_cell,
                                Top_cell, Other}",
    "thickness": "Float [nm]",
    "deposition_procedure": "[{}, {}, ...]",
    "materials_in_layer": "[{}, {}, ...]",
    "post_deposition_procedure": "[{}, {}, ...]",
    "storage_until_use": "{}"
}
```

Other

```
{
    "functionality": "other",
    "functionality_specific": "String",
    "area": "Float [cm²]",
    "layer_index": "Integer",
    "layer_used_as_substrate": "Integer",
    "sheet_resistance": "Float [ohm]",
    "surface_roughness": " Float [nm]",
    "subcell_association": "String {Monolithic_device, Bottom_cell, Middle_cell,
                                Top_cell, Other}",
    "thickness": "Float [nm]",
    "deposition_procedure": "[{}, {}, ...]",
    "materials_in_layer": "[{}, {}, ...]",
    "post_deposition_procedure": "[{}, {}, ...]",
    "storage_until_use": "{}"
}
```

Recombination_layer

```
{
    "functionality": "recombination_layer",
    "functionality_specific": "String",
    "subcell_association": "String {Monolithic_device, Bottom_cell, Middle_cell,
                                Top_cell, Other}",
    "layer_index": "Integer",
    "layer_used_as_substrate": "Integer",
    "area": "Float [cm²]",
    "thickness": "Float [nm]",
    "sheet_resistance": "Float [ohm]",
    "surface_roughness": "Surface roughness [nm]",
    "deposition_procedure": "[{}, {}, ...]",
    "materials_in_layer": "[{}, {}, ...]",
    "post_deposition_procedure": "[{}, {}, ...]",
    "storage_until_use": "{}"
}
```

Refractive index matching

```
{
    "functionality": "refractive_index_matching",
    "area": "Float [cm²]",
```

```
    "layer_index": "Integer",
    "layer_used_as_substrate": "Integer",
    "refractive_index": "Float",
    "sheet_resistance": "Float [ohm]",
    "surface_roughness": "Surface roughness [nm]",
    "subcell_association": "String {Monolithic_device, Bottom_cell, Middle_cell,
                                Top_cell, Other}",
    "thickness": "Float [nm]",
    "deposition_procedure": "[{}, {}, ...]",
    "materials_in_layer": "[{}, {}, ...]",
    "post_deposition_procedure": "[{}, {}, ...]",
    "storage_until_use": "{}"
}
```

Substrate

```
{
    "functionality": "refractive_index_matching",
    "area": "Float [cm²]",
    "layer_index": "Integer",
    "layer_used_as_substrate": "Integer",
    "subcell_association": "String {Monolithic_device, Bottom_cell, Middle_cell,
                                Top_cell, Other}",
    "thickness": "Float [nm]",
    "cleaning_procedure": "[{}, {}, ...]",
    "materials_in_layer": "[{}, {}, ...]",
    "storage_until_use": "{}"
}
```

TCO

```
{
    "functionality": "tco
    "area": "Float [cm²]",
    "layer_index": "Integer",
    "layer_used_as_substrate": "Integer",
    "place_of_deposition": "String {in_house, by_commercial_supplier, other}",
    "sheet_resistance": "Float [ohm]",
    "surface_roughness": "Surface roughness [nm]",
    "subcell_association": "String {Monolithic_device, Bottom_cell, Middle_cell,
                                Top_cell, Other}",
    "thickness": "Float [nm]",
    "cleaning_procedure": " [{}, {}, ...]",
    "deposition_procedure": "[{}, {}, ...]",
    "etching_procedure": " [{}, {}, ...]",
    "materials_in_layer": "[{}, {}, ...]",
    "storage_until_use": "{}"
}
```

Layers photoabsorbers

A bit more attention has been put upon photoabsorber layers, where an additional subdivision has been done by defining subsections for different photoabsorbers. In the graphical user interface, if a "photoabsorber" is selected as the layers functionality, the user get to select which photoabsorber it is, whereupon ley-values and subsection relevant for that photoabsorber is loaded at the same level of nesting.

```
{
    "functionality": "String {photoabsorber}",
    "photoabsorber_material": "String {CIGS, CIS, CZTS, OPV_blend, Organic_dye,
                                       Other, Quantum_dots, Perovskite, Silicon}",
}
```

CIGS

The submodule composition is found under the headline composition_CIGS in the Materials section

```
{
    "functionality": "String {photoabsorber}",
    "photoabsorber_material": "String {CIGS}",
    "area": "Float [cm²]",
    "crystallinity": "String {single_crystal, poly_crystaline, amorphous}",
    "layer_index": "Integer",
    "layer_used_as_substrate": "Integer",
    "sample_origin": "String {lab_made, commercial}",
    "subcell_association": "String {Monolithic_device, Bottom_cell, Middle_cell,
                                     Top_cell, Other}",
    "surface_roughness": " Float [nm]",
    "thickness": "Float [nm]",
    "band_gap": "{}"
    "composition": "{}"
    "deposition_procedure": " [{}, {}, ...]",
    "post_deposition_procedure": "[{}, {}, ...]",
    "secondary_phases_impurities_and_dopants": "[{}, {}, ...]",
    "storage_until_use": "{}"
}
```

CIS

Same fields as for CIGS but with another submodule for composition (composition_CIS)

CZTS

Same fields as for CIGS but with another submodule for composition (composition_CZTS)

OPV blend

```
{
    "functionality": "String {photoabsorber}",
    "photoabsorber_material": "String {OPV_blend}",
    "absorber_type": "String {sigle_layer, bilayer, heterojunction, other}",
    "area": "Float [cm²]",
    "HOMO_level": "Float [eV]",
    "layer_index": "Integer",
    "layer_used_as_substrate": "Integer",
    "LUMO_level": "Float [eV]",
```

```
    "molar_extinction_coefficient": "Float [nm]",
    “peak_absorption_wavelength": "Float [nm]",
    "subcell_association": "String {Monolithic_device, Bottom_cell, Middle_cell,
                                    Top_cell, Other}",
    "thickness": "Float [nm]",
    "band_gap": "{}"
    "chemical_composition": "[{}, {}, ...]",
    "deposition_procedure": " [{}, {}, ...]",
    "post_deposition_procedure": "[{}, {}, ...]",
    "secondary_phases_impurities_and_dopants": "[{}, {}, ...]",
    "storage_until_use": "{}"
}
```

Organic dye

```
{
    "functionality": "String {photoabsorber}",
    "photoabsorber_material": "String {Organic_dye}",
    "area": "Float [cm²]",
    "dye_origin": "String {lab_made, commercial}",
    "HOMO_level": "Float [eV]",
    "layer_index": "Integer",
    "layer_used_as_substrate": "Integer",
    "LUMO_level": "Float [eV]",
    "molar_extinction_coefficient": "Float [nm]",
    "oxidation_potential": "Float [V]",
    “peak_absorption_wavelength": "Float [nm]",
    "subcell_association": "String {Monolithic_device, Bottom_cell, Middle_cell,
                                    Top_cell, Other}",
    "thickness": "Float [nm]",
    "band_gap": "{}"
    "chemical_composition": "[{}, {}, ...]",
    "deposition_procedure": " [{}, {}, ...]",
    "post_deposition_procedure": "[{}, {}, ...]",
    "secondary_phases_impurities_and_dopants": "[{}, {}, ...]",
    "storage_until_use": "{}"
}
```

Other

```
{
    "functionality": "String {photoabsorber}",
    "photoabsorber_material": "String {Other}",
    "area": "Float [cm²]",
    "crystallinity": "String {single_crystal, poly_crystaline, amorphous}",
    "layer_index": "Integer",
    "layer_used_as_substrate": "Integer",
    "sample_origin": "String {lab_made, commercial}",
    "subcell_association": "String {Monolithic_device, Bottom_cell, Middle_cell,
                                    Top_cell, Other}",
    "surface_roughness": "Float [nm]",
    "thickness": "Float [nm]",
```

```
    "band_gap": "{}"
    "chemical_composition": "[{}, {}, ...]",
    "deposition_procedure": " [{}, {}, ...]",
    "post_deposition_procedure": "[{}, {}, ...]",
    "secondary_phases_impurities_and_dopants": "[{}, {}, ...]",
    "storage_until_use": "{}"
}
```

Quantum dots

```
{
    "functionality": "String {photoabsorber}",
    "photoabsorber_material": "String {Quantum_dots}",
    "area": "Float [cm²]",
    "layer_index": "Integer",
    "layer_used_as_substrate": "Integer",
    "particle_size": "Float [nm]",
    "sample_origin": "String {lab_made, commercial}",
    "subcell_association": "String {Monolithic_device, Bottom_cell, Middle_cell,
                                    Top_cell, Other}",
    "thickness": "Float [nm]",
    "band_gap": "{}"
    "chemical_composition": "[{}, {}, ...]",
    "deposition_procedure": "[{}, {}, ...]",
    "ligands": "[{}, {}, ...]",
    "post_deposition_procedure": "[{}, {}, ...]",
    "secondary_phases_impurities_and_dopants": "[{}, {}, ...]",
    "storage_until_use": "{}"
}
```

Perovskite

```
{
    "functionality": "String {photoabsorber}",
    "photoabsorber_material": "String {Perovskite}",
    "area": "Float [cm²]",
    "average_grain_size": "Float [nm]",
    "crystallinity": "String {polycrystalline_film, single_crystal, quantum_dots,
                              nano_rods, amorphous, other}",
    "dimensionality": "String {0D, 1D, 2D, 2D/3D, 3D, other}",
    "has_a_2D_perovskite_caping_layer": "Boolean",
    "is_a_double_perovskite": "Boolean",
    "is_inorganic": "Boolean",
    "is_lead_free": "Boolean",
    "layer_index": "Integer",
    "layer_used_as_substrate": "Integer",
    "sample_origin": "String {lab_made, commercial}",
    "surface_roughness": "Float [nm]",
    "subcell_association": "String {Monolithic_device, Bottom_cell, Middle_cell,
                                    Top_cell, Other}",
    "thickness": "Float [nm]",
    "band_gap": "{}"
    "composition": "[{}, {}, ...]",
```

```
    "deposition_procedure": "[{}, {}, ...]",
    "post_deposition_procedure": "[{}, {}, ...]",
    "secondary_phases_impurities_and_dopants": "[{}, {}, ...]",
    "storage_until_use": "{}"
}
```

silicon

```
{
    "functionality": "String {photoabsorber}",
    "photoabsorber_material": "String {Silicon}",
    "area": "Float [cm²]",
    "cell_type": "String {HJT, PERC, TOPCon, PERL, other}",
    "crystallinity": "String {polycrystalline_film, single_crystal, quantum_dots,
                              nano_rods, amorphous, other}",
    "layer_index": "Integer",
    "layer_used_as_substrate": "Integer",
    "material": "String{Silicon}",
    "subcell_association": "String {Monolithic_device, Bottom_cell, Middle_cell,
                                    Top_cell, Other}",
    "surface_roughness": "Float [nm]",
    "thickness": "Float [nm]",
    "band_gap": "{}"
    "doping_sequence": "{}"
    "deposition_procedure": "[{}, {}, ...]",
    "post_deposition_procedure": "[{}, {}, ...]",
    "secondary_phases_impurities_and_dopants": "[{}, {}, ...]",
    "storage_until_use": "{}"
}
```

Materials

Chemical

The chemical section has one set of fields dealing with naming, and one with extrinsic properties (such as amount, aggregation state, etc.)

```
"chemical":{
    "abbreviation": "String",
    "cas_number": "String",
    "iupac_name": "String",
    "molecular_formula": "String",
    "name": "String",
    "smile": "String",

    "aggregation_state": "String {solid, liquid, gas, solution, other}",
    "is_nanostructured": "Boolean",
    "mass": "Float [g]",
    "number_of_moles": "Float [mol]",
    "partial pressure": "Float [Pa]",
    "purity": "String",
    "volume": "Float [ml]",
    "supplier": "{}",
    "nanostructuration": "{}",
}
```

Chemical composition

Same as “chemical”

Ink

Same as “solution”

Ligands

Same as “chemicals”

Materials in layer

```
"materials_in_layer":{
    "fraction_of_layer_content": "Float []",
    "fraction_of_layer_content_metric": "String {mass_fraction, volume_fraction,
                                                 mol_fraction}",
    "functionality_in_layer": "String {majority_phase, secondary_phase, additive,
                                       dopant, impurity, Other}",

    "abbreviation": "String",
    "cas_number": "String",
    "iupac_name": "String",
    "molecular_formula": "String",
    "name": "String",
    "smile": "String",

    "aggregation_state": "String {solid, liquid, gas, solution, other}",
    "is_nanostructured": "Boolean",
    "mass": "Float [g]",
    "number_of_moles": "Float [mol]",
```

```
    "partial pressure": "Float [Pa]",
    "purity": "String",
    "volume": "Float [ml]",
    "supplier": "{}",
    "nanostructuration": "{}",
}
```

Nanstructuration

```
"nanostructuration":{
    "shape": "String {Nanoparticle, Quantum_dot, Nanorod, Disc, Other}",
    "diameter": "Float [nm]",
    "width": "Float [nm]",
    "length": "Float [nm]",
}
```

Photoabsorber compositions

Composition CIGS

The parameter “long_form” can be generated automatically form the other fields

```
"composition_CIGS":{
    "copper": "Float []",
    "indium": "Float []",
    "gallium": "Float []",
    "selenium": "Float []",
    "long_form": "String",
}
```

Composition CIS

The parameter “long_form” can be generated automatically form the other fields

```
"composition_CIGS":{
    "copper": "Float []",
    "indium": "Float []",
    "selenium": "Float []",
    "long_form": "String",
}
```

Composition CZTS

The parameter “long_form” can be generated automatically form the other fields

```
"composition_CZTS":{
    "copper": "Float []",
    "zink": "Float []",
    "tin": "Float []",
    "sulfur": "Float []",
    "long_form": "String",
}
```

Composition Perovskite

The parameter “long_form” and “short_form” can be generated automatically form the other fields. The subsection a_ions, b_ions, and x_ions are all readin in the subsection “ion” found under the headliine “Perovskite ions”.

```
"composition_Perovskite":{
    "short_form": "String",
```

```
    "long_form": "String",
    "a_ions": "[{}, {}, ...]",
    "b_ions": "[{}, {}, ...]",
    "x_ions": "[{}, {}, ...]",
}
```

Perovskite ions

```
"ion":{
    "abberivation": "String",
    "coefficient": "Float",
    "molecular_formula": "String",
    "smile": "String",
    "common_name": "String",
    "iupac_name": "String",
    "cas_number": "String",
    "source_compound_smile": "String",
    "source_compound_iupac_name": "String",
    "source_compound_cas_number": "String",
}
```

Secondary Phases Impurities and Dopands

Same as "materials in layer"

Solution

```
"solution":{
    "age": "Float [minute]",
    "color": "String",
    "density": "Float [g/ml]",
    "is_filtered": "Boolean",
    "is_stired": "Boolean",
    "lab_id": "String",
    "preparation_date": "Datetime",

    "temperature": "Float [°C]",
    "temperature_max": "Float [°C]",
    "viscosity": "Float [Pa*s]",
    "volume": "Float [ml]",
    "chemicals": "[{}, {}, ...]",
    "environmental_conditions_during_preparation": "{}",
}
```

Supplier

```
"supplier":{
    "place_of_origin": "String {in_house, from_collaborator,
                               from_commercial_supplier}",
    "name_of_supplier": "String",
    "catalog_number": "String",
    "batch_number": "String",
    "delivery_date": "Datetime",
}
```

Target material
Same as “chemical”

Deposition procedure

Depositing a layer may require more than one production step, e.g. a spin-coating, followed by thermal annealing, followed by an ion exchange procedure, followed by another step of thermal annealing. The deposition procedures would therefore naturally be stored in a list of subsequent production steps.

Different deposition procedures require different parameters to describe properly, and thus requires individual data schemas, and we have written schemas for some om the more common deposition procedures.

The layers subsection is calling the deposition procedure submodule under different name. “deposition procedure”, “cleaning procedure”, “etching procedure”, “storage” are all pointing to the same place.

Cleaning procedure

Same entry point as “deposition procedure”

Chemical bath deposition

```
"Chemical_bath_deposition":{
    "date": "Datetime",
    "dipping_time": "Float [minute]",
    "drying_procedure": "String {Gas_blowing, Self_drying, Heating, Tissue_paper,
                                Other}",
    "duration_of_procedure": "Float [s]",
    "number_of_repetitions": "Integer",
    "substrate_temperature": "Float [°C]",
    "time_between_repetitions": "Float [minute]",
    "solution": "{}",
    "environmental_conditions": "{}",
}
```

Dip coating

```
"Dip_coating":{
    "date": "Datetime",
    "dipping_time": "Float [minute]",
    "drying_procedure": "String {Gas_blowing, Self_drying, Heating, Tissue_paper,
                                Other}",
    "duration_of_procedure": "Float [s]",
    "number_of_repetitions": "Integer",
    "substrate_temperature": "Float [°C]",
    "time_between_repetitions": "Float [minute]",
    "solution": "{}",
    "environmental_conditions": "{}",
}
```

Doctorblading

```
"doctorblading":{
    "coater_model": "String",
    "coating_speed": "Float [mm/s]",
    "date": "Datetime",
    "duration_of_procedure": "Float [s]",
    "inc_volume": "Float [ml]",
    "substrate_temperature": "Float [°C]",
```

```
    "time_of_program": "Float [minute]",
    "solution": "{}",
    "environmental_conditions": "{}",
}
```

Etching procedure

Same entry point as “deposition procedure”

Evaporation multi source

```
"Evaporation_multi_source":{
    "chamber_pressure": "Float [Pa]",
    "date": "Datetime",
    "duration_of_procedure": "Float [s]",
    "substrate_rotation_rate": "Float [rpm/s]",
    "substrate_temperature": "Float [°C]",
    "evaporation_sources": "[{}, {}, ...]",
}
```

Evaporation single source

```
"Evaporation_single_source":{
    "chamber_pressure": "Float [Pa]",
    "crucible_material": "String",
    "date": "Datetime",
    "duration_of_procedure": "Float [s]",
    "mass_in_crucible": "Float [g]",
    "substrate_temperature": "Float [°C]",
    "substrate_rotation_rate": "Float [rpm/s]",
    "chemicals": "[{}, {}, ...]",
    "evaporation_segment": "[{}, {}, ...]",
}
```

Heating

```
"Heating":{
    "date": "Datetime",
    "duration_of_procedure": "Float [s]",
    "heating_medium": "String {Hotplate, Furnace, Liquid_bath, Gas, Other}",
    "temperature_segment": "[{}, {}, ...]",
}
```

Inkjet printing

```
"Inkjet_printing":{
    "date": "Datetime",
    "drop_size": "Float [minute]",
    "duration_of_procedure": "Float [s]",
    "ink_volume": "Float [ml]",
    "printer_model": "String",
    "substrate_temperature": "Float [°C]",
    "solution": "{}",
    "environmental_conditions": "{}",
}
```

Ion exchange by dipping

```
"Ion_exchange_by_dipping":{
    "date": "Datetime",
    "dipping_time": "Float [minute]",
    "drying_procedure": "String {Gas_blowing, Self_drying, Heating, Tissue_paper,
                                Other}",
    "duration_of_procedure": "Float [s]",
    "number_of_repetitions": "Integer",
    "substrate_temperature": "Float [°C]",
    "time_between_repetitions": "Float [minute]",
    "solution": "{}",
    "environmental_conditions": "{}",
}
```

Ion exchange by gas diffusion

```
"Ion_exchange_by_gas_diffusion":{
    "date": "Datetime",
    "duration_of_procedure": "Float [s]",
    "exposure_time": "Float [minute]",
    "pressure_of_reaction_mixture": "Float [Pa]",
    "substrate_temperature": "Float [°C]",
    "temperature_of_reaction_mixture": "Float [°C]",
    "chemicals": "[{}, {}, ...]",
    "environmental_conditions": "{}",
}
```

Post deposition procedure

Same entry point as “deposition procedure”

Slot dye coating

```
"Slot_dye_coating":{
    "coater_model": "String",
    "coating_speed": "Float [mm/s]",
    "date": "Datetime",
    "duration_of_procedure": "Float [s]",
    "ink_volume": "Float [ml]",
    "substrate_temperature": "Float [°C]",
    "time_between_repetitions": "Float [minute]",
    "solution": "{}",
    "environmental_conditions": "{}",
}
```

Spin-coating

```
"Spin_coating":{
    "date": "Datetime",
    "dispense_speed": "Float [ml/s]",
    "distance_between_tip_and_substrate": "Float [mm]",
    "duration_of_procedure": "Float [s]",
    "is_antisolvent_quenching_used": "Boolean",
    "is_gas_quenching_used": "Boolean",
    "solvent_volume": "Float [ml]",
```

```
    "spin-coater_model": "String",
    "static_or_dynamic_spin-coating": "String {Static, dynamic}",
    "substrate_temperature": "Float [°C]",
    "time_at_dispense_start": "Float [s]",
    "time_of_program": "Float [s]",
    "antisolvent_details": "{}",
    "environmental_conditions": "{}",
    "gas_quenching_details": "{}",
    "solution": "{}",
    "spin_coating_segments": "[{}, {}, ...]",
}
```

Spray-coating

```
"Spray_coating":{
    "carrier_gas": "String {Air, Dry_air, N2, Ar, O2, He, Other}",
    "coater_model": "String",
    "date": "Datetime",
    "deposition_area": "Float [ml]",
    "duration_of_procedure": "Float [s]",
    "gas_pressure": "Float [Pa]",
    "ink_volume": "Float [ml]",
    "nozzle_sample_distance": "Float [cm]",
    "nozzle_diameter": "Float [mm]",
    "substrate_temperature": "Float [°C]",
    "time_of_program": "Float [s]",
    "environmental_conditions": "{}",
    "ink": "{}",
}
```

Sputtering

```
"Sputtering":{
    "chamber_pressure": "Float [Pa]",
    "date": "Datetime",
    "duration_of_procedure": "Float [s]",
    "substrate_rotation_rate": "Float [rpm/s]",
    "substrate_temperature": "Float [°C]",
    "environmental_conditions": "{}",
    "sputtering-segment": "[{}, {}, ...]",
    "target_material": "[{}, {}, ...]",
}
```

Storage

```
"Storage":{
    "date": "Datetime",
    "duration_of_procedure": "Float [s]",
    "environmental_conditions": "{}",
}
```

Storage until use

Same as “storage”

Deposition procedure – sub sections

Antisolvent details

```
"Antisolvent_details":{
    "dispense_speed": "Float [ml/s]",
    "distance_between_tip_and_substrate": "Float [mm]",
    "preparation_date": "Datetime",
    "time_at_dispense_start": "Float [s]",
    "volume": "Float [ml]",
    "solution": "{}",
}
```

Environmental conditions

```
"environmental_conditions":{
    "ambient_temperature": "Float [°C]",
    "drying_procedure": "String {Air, Dry_air, N2, Ar, Vacuum, O2, He, Other}",
    "in_glove_box": "Boolean",
    "oxygen_concentration": "Float",
    "pressure": "Float [Pa]",
    "relative_humidity": "Float",
}
```

Evaporation source

```
"Evaporation_source":{
    "crucible_material": "String",
    "mass_in_crucible": "Float [g]",
    "chemicals": "[{}, {}, ...]",
    "evaporation_segment": "[{}, {}, ...]",
}
```

Evaporation segment

```
"Evaporation_segment":{
    "deposition_rate": "Float [nm/s]",
    "evaporation_rate": "Float [g/s]",
    "time_of_segment": "Float [s]",
}
```

Gas quenching details

```
"Gas_quenching_details":{
    "duration": "Float [s]",
    "gas": "String {Air, Dry_air, N2, Ar, O2, He, Other}",
    "pressure": "Float [Pa]",
    "staring_time": "Float [s]",
}
```

Spin-coating segments

```
"Spin-coating_segments":{
    "acceleration": "Float [rpm/s]",
    "rpm_at_start": "Float [rpm]",
    "rpm_at_end": "Float [rpm]",
    "time_of_segment": "Float [s]",
}
```

Sputtering segments

```
"sputtering_segments":{
    "deposition_rate": "Float [nm/s]",
    "target_power": "Float [J/s]",
    "time_of_segment": "Float [s]",
}
```

Temperature segments

```
"temperature_segments":{
    "temperature_at_start": "Float [°C]",
    "temperature_at_end": "Float [°C]",
    "temperature_acceleration": "Float [°C/minute]",
    "time_of_segment": "Float [s]",
}
```

Measurements

Measurements - Top level

```
"measurements":{
    "EQE": "List of measurements [{}, {}, ...]",
    "Flexibility": "List of measurements [{}, {}, ...]",
    "JV-measurements": "List of measurements [{}, {}, ...]",
    "Outdoor_performance": "List of measurements [{}, {}, ...]",
    "Stability": "List of measurements [{}, {}, ...]",
    "Stabilised_performance": "List of measurements [{}, {}, ...]",
    "Transmission": "List of measurements [{}, {}, ...]"
}
```

EQE

```
"EQE":{
    "age_of_cell": "Float [hr]",
    "cell_temperature": "Float [°C]",
    "date": "Datetime",
    "equipment_brand_name": "String",
    "illuminated_area": "Float [cm²]",
    "integration_time": "Float [s]",
    "is_under_bias_light": "Boolean",
    "measurement_done_on": "String {Compleat_device, Bottom_cell, Middle_cell,
                                    Top_cell, shaded_top_cell, shaded_middle_cell,
                                    Other}",
    "measurement_time": "Float [s]",
    "scan_speed": "Float [nm/s]",
    "step_size": "Float [nm]",
    "wavelength_at_start": "Float [nm]",
    "wavelength_at_end": "Float [nm]",
    "bias_light": "{}",
    "environmental_conditions": "{}",
    "EQE_metrics": "{}",
    "raw_data": "{}",
}
```

Flexibility

```
"Flexibility":{
    "age_of_cell": "Float [hr]",
    "bending_radius": "Float [°]",

    "date": "Datetime",
    "measurement_done_on": "String {Compleat_device, Bottom_cell, Middle_cell,
                                    Top_cell, shaded_top_cell, shaded_middle_cell,
                                    Other}",
    "number_of_bending_cycles": "Integer",
    "PCE_after_bending": "Float",
    "PCE_before_bending": "Float",
    "environmental_conditions": "{}",}
```

JV measurement

```
"JV_measurement":{
    "age_of_cell": "Float [hr]",
    "cell_temperature": "Float [°C]",
    "date": "Datetime",
    "delay_time": "Float [s]",
    "integration_time": "Float [s]",
    "is_certified": "Boolean",
    "is_preconditioned_before_measurement": "Boolean",
    "measurement_done_on": "String {Compleat_device, Bottom_cell, Middle_cell,
                                    Top_cell, shaded_top_cell, shaded_middle_cell,
                                    Other}",
    "scan_speed": "Float [V/s]",
    "source_meter_brand_name": "String",
    "voltage_step": "Float [mV]",
    "certification_details": "{}",
    "environmental_conditions": "{}",
    "jv_metrics": "{}",
    "light_conditions": "{}",
    "preconditioning": "{}",
    "raw_data": "{}",
}
```

Outdoor performance

```
"Outdoor_performance": {
    "bias_current": "Float [mA/cm²]",
    "bias_potential": "Float [V]",
    "bias_potential_type": "String {MPPT, Open_circuit, Constant_potential,
                                    Constant_current, Passive_resistance, Other}",
    "bias_resistance": "Float [ohm]",
    "is_JV_curves_measured_periodically": "Boolean",
    "location_city": "String",
    "location_latitude": "Float",
    "location_longitude": "Float",
    "tilt": "Float [°]",
    "time_start": "Datetime",
    "time_end": "Datetime",

    "max_humidity": "Float",
    "max_ambient_temperature": "Float [°C]",
    "min_ambient_temperature": "Float [°C]",
    "max_cell_temperature": "Float [°C]",
    "min_cell_temperature": "Float [°C]",
    "measurement_protocol": "String {ISOS-O-1', ISOS-O-2, ISOS-O-3, Other}",
    "number_of_solar_tracking_axes": "Integer",
    "certification_details": "{}",
    "JV-measurements:": "[{}, {}, ...]",
    "raw_data": "{}",
    "stability_metrics": "{}",
}
```

Stabilised performance

```
"Stabilised_performance":{
    "age_of_cell": "Float [hr]",
    "cell_temperature": "Float [°C]",
    "current": "Float [mA/cm²]",
    "date": "Datetime",
    "measurement_done_on": "String {Compleat_device, Bottom_cell, Middle_cell,
                                    Top_cell, shaded_top_cell, shaded_middle_cell,
                                    Other}",
    "potential": "Float [V]",
    "stabilisation_time": "Float [s]",
    "type_of_measurement": "String {Constant_potential, Constant_current,
                                    Maximum_powerpoint_tracking}",
    "certification_details": "{}",
    "environmental_conditions": "{}",
    "light_conditions": "{}",
    "raw_data": "{}",
    "stabilised_performance_metrics": "{}",

}
```

Stability

```
"Stability":{
    "age_of_cell": "Float [hr]",
    "cell_temperature": "Float [°C]",
    "bias_current": "Float [mA/cm²]",
    "bias_potential_type": "String {MPPT, Open_circuit, Constant_potential,
                                    Constant_current, Passive_resistance, Other}",
    "bias_potential": "Float [V]",
    "bias_resistance": "Float [ohm]",

    "date": "Datetime",
    "is_certified": "Boolean",
    "is_illuminated": "Boolean",
    "is_JV_curves_measured_periodically": "Boolean",
    "is_measured_with_load_cycling": "Boolean",
    "length_of_measurement": "Float [hr]",
    "measurement_done_on": "String {Compleat_device, Bottom_cell, Middle_cell,
                                    Top_cell, shaded_top_cell, shaded_middle_cell,
                                    Other}",
    "measurement_protocoll": "String {ISOS-D-1, ISOS-D-1l, ISOS-D-2, ISOS-D-2l,
                                    ISOS-D-3, ISOS-V-1, ISOS-V-1l, ISOS-V-2, ISOS-V-
                                    2l, ISOS-V-3, ISOS-L-1, ISOS-L-1l, ISOS-L-2, ISOS-
                                    L-2l, ISOS-L-3, ISOS-O-1, ISOS-O-2, ISOS-O-3,
                                    ISOS-T-1, ISOS-T-1l, ISOS-T-2, ISOS-T-2l, ISOS-T-
                                    3, ISOS-T-3l, ISOS-LC-1, ISOS-LC-1l, ISOS-LC-2,
                                    ISOS-LC-2l, ISOS-LC-3, ISOS-LC-3l, ISOS-LT-1,
                                    ISOS-LT-2, ISOS-LT-3, IEC 61215, Other}",
    "certification_details": "{}",
    "environmental_conditions": "{}",
```

```
    "JV-measurements": "[{}, {}, ...]",
    "light_conditions": "{}",
    "load_cycling": "[{}, {}, ...]",
    "raw_data": "{}",
    "stability_metrics": "{}",
}
```

Transmission

```
"Transmission":{
    "age_of_cell": "Float [hr]",
    "average_transmission_in_the_visible_range": "Float",
    "date": "Datetime",
    "measurement_done_on": "String {Compleat_device, Bottom_cell, Middle_cell,
                                    Top_cell, shaded_top_cell, shaded_middle_cell,
                                    Other}",
    "spectrometer_model": "String",
    "raw_data": "{}",
}
```

Measurements – Sub sections

Certified performance

```
"Certified_performance":{
    "is_certified": "Boolean",
    "certification_date": "Datetime",
    "certification_institute": "String",
    "certification_protocol": "String",
    "ff": "Float [j_mp*v_mp/(j_sc*voc)]",
    "is_certified": "Boolean",
    "j_sc": "Float [mA/cm²]",
    "j_mp": "Float [mA/cm²]"
    "pce_stabilized": "Float",
    "pce_from_iv_data": "Float",
    "v_mp": "Float [V]",
    "voc": "Float [V]",
}
```

Certification details

```
"Certification_details":{
    "certification_date": "Datetime",
    "certification_institute": "String",
    "certification_protocol": "String",
    "is_certified": "Boolean",
}
```

EQE metrics

```
"EQE_metrics":{
    "EQE_inflection_point": "Float [nm]",
    "EQE_max": "Float []",
    "EQE_onset": "Float [nm]",
    "integrated_current": "Float [mA/cm²]",
}
```

JV metrics

```
"JV_metrics": {
    "ff": "Float [j_mp*v_mp/(j_sc*voc)]",
    "is_certified": "Boolean",
    "j_sc": "Float [mA/cm²]",
    "j_mp": "Float [mA/cm²]"
    "pce": "Float",
    "rs": "Float [ohm]",
    "rsh": "Float [ohm]",
    "v_mp": "Float [V]",
    "voc": "Float [V]",
}
```

Light conditions

```
"Light_conditions": {
    "illumination": "Float [lux]",
    "illuminated_area": "Float [cm²]",
```

```
    "light_intensity": "Float [mW/cm²]",
    "light_source": "String {Metal halide, Sulfur plasma, LED, White LED, Laser,
                            Xenon-plasma, Generic solar simulator, UV-lamp,
                            Ambient outdoor, Ambient indoor, Dark_conditions,
                            Other}",
    "light_source_brand_name": "String",
    "light_spectra": "String {AM 1.5, AM 1.0, AM 0, UVA, UVB, Monochromatic,
                             Ambient outdoor, Ambient indoor, Dark, Other}",
    "peak_wavelength": "Float [nm]",
    "shadow_mask_is_used": "Boolean",
    "solar_simulator_class": "String {AAA, AAB, AAC, ABA, ABB, ABC, ACA, ACB, ACC,
                                      BAA, BAB, BAC, BBA, BBB, BBC, BCA, BCB, BCC,
                                      CAA, CAB, CAC, CBA, CBB, CBC, CCA, CCB, CCC,
                                       Unknown}",
}
```

Load cycling

```
"Load_cycling": {
    "ambient_temperature_on": "Float [°C]",
    "ambient_temperature_off": "Float [°C]",
    "bias_current_on": "Float [mA/cm²]",
    "bias_current_off": "Float [mA/cm²]",
    "bias_resistance_on": "Float [ohm]",
    "bias_resistance_off": "Float [ohm]",
    "cell_temperature_on": "Float [°C]",
    "cell_temperature_off": "Float [°C]",
    "light_intensity_on": "Float [mW/cm²]",
    "light_intensity_off": "Float [mW/cm²]",
    "number_of_cycles": "Integer",
    "potential_bias_type_on": "String {MPPT, Open_circuit, Constant_potential,
                                       Constant_current, Passive_resistance,
                                       Other}",
    "potential_bias_type_off": " String {MPPT, Open_circuit, Constant_potential,
                                       Constant_current, Passive_resistance,
                                       Other}",
    "potential_bias_on": "Float [V]",
    "potential_bias_off": "Float [V]",
    "relative_humidity_on": "Float",
    "relative_humidity_off": "Float",
    "time_on": "Float [minute]",
    "time_off": "Float [minute]",
}
```

Outdoor metrics

```
"Outdoor_metrics": {
    "average_power_generated": "Float [kWh/year/m²]",
    "burn_in_period_observed": "Boolean",
    "burn_in_period_length": "Float [hr]",
    "PCE_at_start": "Float",
    "PCE_at_end": "Float",
```

```
    "PCE_1000h": "Float",
    "total_power_generated": "Float [kWh/m²]",

    "T80": "Float [hr]",
    "T80s": "Float [hr]",
    "T95": "Float [hr]",
    "T95s": "Float [hr]",
}
```

Preconditioning

```
"Preconditioning": {
    "time": "Float [minute]",
    "potential": "Float [V]",
    "light_intensity": "Float [mW/cm²]",
}
```

Stabilised performance metrics

```
"Stabilised_performance_metrics": {
    "pce": "Float",
    "v_mp": "Float [V]",
    "j_mp": "Float [mW/cm²]",
}
```

Stability metrics

```
"Stability_metrics": {
    "burn_in_period_observed": "Boolean",
    "burn_in_period_length": "Float [hr]",
    "PCE_at_start": "Float",
    "PCE_at_end": "Float",
    "PCE_1000h": "Float",
    "T80": "Float [hr]",
    "T80s": "Float [hr]",
    "T95": "Float [hr]",
    "T95s": "Float [hr]",
}
```

Measurements - Raw data

Raw data eqe

```
"Raw_data_eqe": {
    "wavelength": "['*'] Float [nm]",
    "EQE": "['*'] Float",
    "time": "['*'] Float [s]",
}
```

Raw data jv

```
"Raw_data_jv": {
    "potential": "['*'] Float [V]",
    "current": "['*'] Float [mA/cm²]",
}
```

Raw data outdoor performance

```
"Raw_data_outdoor_performance": {
    "time": "['*'] Float [s]",
    "pce": "['*']",
    "potential": "['*'] Float [V]",
    "current": "['*'] Float [mA/cm²]",
    "light_intensity": "['*'] Float [mW/cm²]",

}
```

Raw data stabilized performance

```
"Raw_data_stabilised_performance": {
    "time": "['*'] Float [s]",
    "pce": "['*']",
    "potential": "['*'] Float [V]",
    "current": "['*'] Float [mA/cm²]",
}
```

Raw data stability

```
"Raw_data_stability": {
    "time": "['*'] Float [s]",
    "pce": "['*']",
    "potential": "['*'] Float [V]",
    "current": "['*'] Float [mA/cm²]",
}
```

Raw data transmission

```
"Raw_data_transmission": {
    "wavelength": "['*'] Float [nm]",
    "transmittance": "['*'] Float",
    "reflectance": "['*'] Float",
    "absorbance": "['*'] Float",
}
```

Module Data

Module data – top level

```
"Module_data":{
    "contains_subcell_that_is_a_module": "Boolean",
    "is_a_monolithic_module": "Boolean",
    "module_description": "[{}, {}, ...]",
}
```

Module description

```
"Module_description":{
    "active_module_area": "Float [cm²]",
    "individual_cell_area": "Float [cm²]",
    "number_of_cells": "Integer",
    "subcell_association": " String {Compleat_device, Bottom_cell, Middle_cell,
                                Top_cell, shaded_top_cell, shaded_middle_cell,
                                Other}",
    "total_module_area": "Float [cm²]",
}
```